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github.com/eycoo

Education

Institut Teknologi Sepuluh Nopember

Jul. 2024 – Present

Bachelor of Science in Artificial Intelligence Engineering (GPA: 3.4/4.00)

Surabaya, Indonesia

Experience

MengAsihi RAG Based AI Chatbot

May. 2026 – Present

Freelance AI Engineer

Surabaya, Indonesia

- Designing a RAG pipeline for a breastfeeding guidance chatbot, including selection of LLM, vector database, and retrieval strategy suitable for a health-sensitive domain.
- Planning a microservice backend with REST API and native Android frontend, with attention to inference latency and response guardrails to reduce hallucination risks.

Avalon AI Corporation

Apr. 2026 – Present

AI Research Assistant

Surabaya, Indonesia

- Working on proprietary foundational model research under NDA, in the area of neuromorphic vision and spike-based image classification using Izhikevich neuron dynamics and temporal encoding.
- Built a custom SNN training pipeline in pure PyTorch, including custom layers, surrogate gradient functions, and temporal encoders, without using existing SNN frameworks.

Schematics

Feb. 2026 – Present

Expert Staff of Data Science Sub Division

Surabaya, Indonesia

- Joined as early staff in Schematics' newly formed Data Science competition division, helping define the technical structure and standards of the division.
- Reviewed and validated competition datasets to check data quality, integrity, and suitability for use in a competitive setting.

Innovillage

Jan. 2026 – April. 2026

AI Engineer

Sidoarjo, Indonesia

- Received funding to build a digital village solution for optimizing Tilapia fish farming.
- Built a Computer Vision model using Custom CNNs with CBAM attention modules to detect fish diseases and classify fish health status.

Pusat Studi Kecerdasan Artifisial dan Teknologi Digital

Jan. 2026 – Feb. 2026

AI Research Intern

Surabaya, Indonesia

- Built an EEG signal analysis pipeline to classify ALS vs. Normal subjects from 1,809 EDF recordings (6 ALS, 150 Normal subjects) across 9 task scenarios.
- Built a chunking and feature extraction pipeline computing time-domain features (MAV, Variance, STD) per subband (Delta, Theta, Alpha, Beta, Gamma) from segmented EEG tasks (Resting, Thinking, Typing).

REXTRA

Aug. 2025 – Feb. 2026

Backend AI Engineer

Surabaya, Indonesia

- Maintained REST API integration for LLM inference calls, handling communication between the backend service and external model providers.
- Monitored API response handling, error management, and request latency to keep LLM-powered features running reliably.

ITS Challenge

Jun. 2025 – Sept. 2025

Staff of Data Center

Surabaya, Indonesia

- Managed collection, validation, and integrity of participant databases for university-scale events.
- Organized data workflows using spreadsheets to support efficient information retrieval.

INCLENATION

Jun. 2025 – Aug. 2025

Staff of Data Management

Surabaya, Indonesia

- Managed collection, validation, and integrity of participant databases for university-scale events.
- Organized data workflows using spreadsheets to support efficient information retrieval.

Projects

Medical Chest X-Ray Classification via Stable Diffusion | *Computer Vision* 2026

- Classified chest X-ray images into 6 categories using a frozen Stable Diffusion U-Net as a feature extractor, combined with a Dual Feature Aggregation module.
- Outperformed DINOv2, MaxViT, and ConvNeXtV2 baselines across 4 scenarios; best results were F1-Macro of 45.17% and AUC-OVR of 77.47% on balanced data with feature-space augmentation, and F1-Macro of 38.55% on imbalanced data without augmentation.

FinVibe | *NLP, Computer Vision, System Design* 2026

- Led a 4-person team building a WhatsApp-based AI system that converts unstructured UMKM transaction data into SAK EMKM-compliant financial reports, credit scores, and RAG-based advisory.
- Designed a microservice architecture using IndoBERT for NER, Whisper for ASR, YOLOv8 Nano and PaddleOCR for receipt parsing, and Phi-3 Mini with pgvector for retrieval-augmented generation.

Synthetic Data for Indonesian Ethnicities | *Stable Diffusion XL, LoRA* 2025

- Built a synthetic data generation pipeline using Stable Diffusion XL to address class imbalance in Indonesian tribal face datasets.
- Fine-tuned the model with Low-Rank Adaptation (LoRA) to generate images that preserve distinct ethnic facial features.
- Showed that adding synthetic data to the training set improved accuracy of external classifiers (XGBoost/ViT), supporting the use of synthetic data in low-resource settings.

Sequential vs Frequency Approach on Audio Classification | *HMM, Naive Bayes, Audio Processing* 2025

- Compared sequential (HMM) and frequency-based (Multinomial Naive Bayes) approaches for audio classification using MFCC feature extraction and K-Means-based Vector Quantization.
- Tested both models across 3 scenarios: codebook size, training data size, and audio duration; Naive Bayes reached up to 90% accuracy and 95% precision at full training data.
- Found that Naive Bayes performed better on frequency-dominant tasks, while HMM was more competitive at smaller codebook sizes (85% accuracy at 32 clusters).

TB Cough Detection System | *Deep Learning, LSTM, Audio Processing* 2025

- Gold Medal, Software Development Division – Gemastik XVIII:** Built an audio-based diagnosis system to detect Tuberculosis from cough recordings.
- Extracted features using MFCCs and spectral descriptors to capture patterns in respiratory audio.
- Trained an LSTM network on sequential audio data, with results showing good sensitivity in separating TB coughs from healthy subjects.

CodeSight | *Computer Vision, YOLOv8, OpenCV* 2025

- Built a learning tool for visually impaired students that converts physical flowchart blocks into executable code using Computer Vision.
- Trained a custom YOLO model for real-time object detection under varied lighting and block orientations.
- Provided audio feedback so students can learn programming logic through physical block arrangements without screen dependency.

Cough Pose Detection & DoA | *Computer Vision, Audio Processing* 2025

- Built a system combining Direction of Arrival (DoA) audio tracking with visual pose detection to identify coughing individuals.
- Designed the algorithm to locate and confirm which person in a group is coughing based on both audio and visual signals.

Honors & Awards

Gold Medal, Software Development Division – Gemastik (2025)

Top 50, Social Project Program Innovillage (2025)

1st Place, BRIN AIDeaNation (2025)

Bronze Medal, WISPO – World Innovative Science Project Olympiad (2023)

Technical Skills

Languages: Python, SQL, C++

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, Pandas, OpenCV, HuggingFace Transformers, LangChain, FastAPI

Developer Tools: Git, VS Code, Google Colab, Kaggle, REST API

Soft Skills: Problem Solving, Teamwork, Time Management